Solutions to Exercise 13A

Chapter 13: Algebraic Expressions

Overview of Key Concepts

Before diving into the exercises, we recall the foundational definitions and rules from the chapter:

- **Algebraic Expression:** A combination of constants and variables connected by the mathematical operators ($+$, $-$, $\times$, $\div$).
- **Monomial:** An algebraic expression containing exactly **one term**. For example, $3x^2 \div p = \frac{3x^2}{p}$ is a monomial.
- **Binomial:** An algebraic expression containing exactly **two terms**. For example, $5 + 3x^3y^3z^2$ is a binomial.
- **Trinomial:** An algebraic expression containing exactly **three terms**. For example, $xy + yz - zx$ is a trinomial.
- **Like Terms:** Terms that have the same literal coefficients (variables raised to the same powers). For example, $-2xy^2$ and $5y^2x$ are like terms.
- **Polynomial:** An algebraic expression in which the variables involved have only **non-negative integral powers**. The **degree** of a polynomial is the highest power of its variables (or sum of powers of variables in a multi-variable term).

Question 1: Term Classification

Identify monomials, binomials and trinomials from the following algebraic expressions:

(i) $7p^2 \times a^3b$

- *Explanation:* The expression can be simplified as $7p^2a^3b$ (or $7a^3bp^2$). Since the variables are multiplied and not separated by addition or subtraction, it consists of only **one term**.
- **Classification: Monomial**

(ii) $5 + 3x^3y^3z^2$

- *Explanation:* This expression consists of two terms: 5 and $3x^3y^3z^2$, which are connected by a $+$ sign.
- **Classification: Binomial**

(iii) $3x^2 \div p$

- *Explanation:* The expression can be written as $\frac{3x^2}{p}$. Since division does not split expressions into separate terms, this constitutes a **single term**.
- **Classification: Monomial**

(iv) $\frac{3a+2b-5c}{7}$

- *Explanation:* This expression can be expanded as $\frac{3}{7}a + \frac{2}{7}b - \frac{5}{7}c$. It has three distinct terms separated by $+$ and $-$ signs.

- **Classification: Trinomial**

(v) $xy + yz - zx$

- *Explanation:* This expression consists of three distinct terms: xy, yz, and $-zx$, separated by $+$ and $-$ signs.
- **Classification: Trinomial**

(vi) $ax^2 + bx \times y^2$

- *Explanation:* This can be simplified as $ax^2 + bxy^2$. It consists of two terms (ax^2 and bxy^2) separated by a $+$ sign.
- **Classification: Binomial**

Question 2: Coefficients of Monomials

Write down the numerical as well as literal coefficient of each of the following monomials:

Let's summarize the numerical and literal coefficients for each of the given monomials:

Part	Monomial	Numerical Coefficient	Literal Coefficient
(i)	$-2p^3q^2$	-2	p^3q^2
(ii)	$\frac{5}{9}xyz^2$	$\frac{5}{9}$	xyz^2
(iii)	$-\frac{3ab^2}{2}$	$-\frac{3}{2}$	ab^2
(iv)	$-\frac{3}{a^2}$	-3	$\frac{1}{a^2}$ (or a^{-2})
(v)	$\frac{2ab}{c}$	2	$\frac{ab}{c}$
(vi)	$-\frac{2x^2}{yz}$	-2	$\frac{x^2}{yz}$

Table 1: Numerical and Literal Coefficients of Monomials

Question 3: Finding Specific Coefficients

In $-5p^2q^3r^4$, write down the coefficient of:

Rule: The coefficient of a given factor is the product of all the remaining factors in that term.

(i) p^2
Since $-5p^2q^3r^4 = (p^2) \times (-5q^3r^4)$,
the coefficient of p^2 is $-5q^3r^4$.

(ii) $5pq^2$
Since $-5p^2q^3r^4 = (5pq^2) \times (-pqr^4)$,
the coefficient of $5pq^2$ is $-pqr^4$.

(iii) p^2q^2
Since $-5p^2q^3r^4 = (p^2q^2) \times (-5qr^4)$,
the coefficient of p^2q^2 is $-5qr^4$.

(iv) pqr
Since $-5p^2q^3r^4 = (pqr) \times (-5pq^2r^3)$,
the coefficient of pqr is $-5pq^2r^3$.

(v) $-pq^2r^3$
Since $-5p^2q^3r^4 = (-pq^2r^3) \times (5pqr)$,
the coefficient of $-pq^2r^3$ is $5pqr$.

(vi) $5q^3r$
Since $-5p^2q^3r^4 = (5q^3r) \times (-p^2r^3)$,
the coefficient of $5q^3r$ is $-p^2r^3$.

Question 4: Identifying Like Terms

Identify like terms in the following:

(i) $a^2, b^2, -2a^2, c^2, 4a$

- *Analysis:* The literal coefficients are a^2, b^2, a^2, c^2, and a respectively.
- **Like Terms:** a^2 and $-2a^2$ (both have the literal coefficient a^2).

(ii) $3x, 4xy, -yz, \frac{1}{2}zy$

- *Analysis:* By commutative law of multiplication, $zy = yz$. Thus, the terms $-yz$ and $\frac{1}{2}zy$ have the same literal coefficient yz.
- **Like Terms:** $-yz$ and $\frac{1}{2}zy$

(iii) $-2xy^2, x^2y, 5y^2x, x^2z$

- *Analysis:* By commutative law, $y^2x = xy^2$. Thus, $-2xy^2$ and $5y^2x$ have the same literal coefficient xy^2.
- **Like Terms:** $-2xy^2$ and $5y^2x$

(iv) $abc, ab^2c, acb^2, c^2ab, b^2ac, a^2bc, cab^2$

- *Analysis:* Let's list and rearrange the variables in alphabetical order for clarity:
  - $abc \implies abc$
  - $ab^2c \implies ab^2c$
  - $acb^2 \implies ab^2c$
  - $c^2ab \implies abc^2$
  - $b^2ac \implies ab^2c$
  - $a^2bc \implies a^2bc$
  - $cab^2 \implies ab^2c$
- **Like Terms:** ab^2c, acb^2, b^2ac, and cab^2 (all simplify to the literal coefficient ab^2c).

Question 5: Generating Algebraic Expressions

Generate algebraic expressions for the following:

(i) The product of a and b added to their sum.

- *Translation:* Product $= ab$, Sum $= a + b$. Added together:
- **Expression:** $(a + b) + ab$ (or $a + b + ab$)

(ii) The quotient of x by 8 is multiplied by y.

- *Translation:* Quotient $= \frac{x}{8}$, multiplied by y:
- **Expression:** $\frac{x}{8} \cdot y = \frac{xy}{8}$

(iii) Thrice x added to y squared.

- *Translation:* Thrice $x = 3x$, y squared $= y^2$. Added together:
- **Expression:** $y^2 + 3x$

(iv) One-third of x multiplied by the sum of p and q.

- *Translation:* One-third of $x = \frac{1}{3}x$, Sum of p and $q = p + q$. Multiplied:
- **Expression:** $\frac{1}{3}x(p+q)$ (or $\frac{x(p+q)}{3}$)

(v) The number to be added to $x + y$ to make it equal to p.

- *Translation:* Let the number be N. We have $(x + y) + N = p \implies N = p - (x + y)$.
- **Expression:** $p - (x + y)$ (or $p - x - y$)

(vi) From a rod $(p + q)$ units in length, n equal pieces are cut. Find the length of each piece.

- *Translation:* Total length $= p + q$. Number of pieces $= n$. Length per piece:
- **Expression:** $\frac{p+q}{n}$

(vii) Five times m subtracted from the sum of p and thrice x.

- *Translation:* Five times $m = 5m$, Sum of p and thrice $x = p + 3x$. Subtracted from the sum:
- **Expression:** $(p + 3x) - 5m$

(viii) The number obtained when m times the difference of x and y is subtracted from n times the sum of x and y.

- *Translation:* m times difference $= m(x - y)$, n times sum $= n(x + y)$. Subtracted:
- **Expression:** $n(x + y) - m(x - y)$

(ix) The sum of 7 numbers each equal to p.

- *Translation:* p added to itself 7 times.
- **Expression:** $7p$

(x) The product of three numbers a, b and c subtracted from the sum of x and y.

- *Translation:* Product $= abc$, Sum $= x + y$. Subtracted from the sum:
- **Expression:** $(x + y) - abc$

Question 6: Polynomials and Degrees

Identify which of the following expressions are polynomials. If so, write their degrees:

(i) $\frac{2}{3}x^4 - 7x + 15$

- *Analysis:* The variable x has powers of 4 and 1, which are non-negative integers.
- **Is a Polynomial: Yes**
- **Degree:** 4 (highest power of x)

(ii) $8x^2 - 3x + 6\sqrt{x} + 1$

- *Analysis:* The term $6\sqrt{x}$ can be written as $6x^{1/2}$. The power of x is $\frac{1}{2}$, which is not a non-negative integer.
- **Is a Polynomial: No**

(iii) $5x^2 - \frac{2}{11}x + 7$

- *Analysis:* The variable x has powers of 2 and 1, which are non-negative integers.
- **Is a Polynomial: Yes**
- **Degree:** 2

(iv) $9x^2y^2 - 3xy^2 + 5x^2y - 6x$

- *Analysis:* All variables have non-negative integer powers. Let's check the degrees of each term:
  - $9x^2y^2$: degree $= 2 + 2 = 4$
  - $-3xy^2$: degree $= 1 + 2 = 3$
  - $5x^2y$: degree $= 2 + 1 = 3$
  - $-6x$: degree $= 1$
- **Is a Polynomial: Yes**
- **Degree:** 4 (highest term degree, from $9x^2y^2$)

(v) $6p^4 - p^3q^2 + pq^3 + q^4$

- *Analysis:* All terms have non-negative integer powers. Let's check the degrees of each term:
  - $6p^4$: degree $= 4$
  - $-p^3q^2$: degree $= 3 + 2 = 5$
  - pq^3: degree $= 1 + 3 = 4$
  - q^4: degree $= 4$
- **Is a Polynomial: Yes**
- **Degree:** 5 (highest term degree, from $-p^3q^2$)

(vi) $4x^5 - 7x^5y + 3xy^4 + 8y^5$

- *Analysis:* All terms have non-negative integer powers. Let's check the degrees of each term:
  - $4x^5$: degree $= 5$
  - $-7x^5y$: degree $= 5 + 1 = 6$
  - $3xy^4$: degree $= 1 + 4 = 5$
  - $8y^5$: degree $= 5$

- **Is a Polynomial: Yes**
- **Degree:** 6 (highest term degree, from $-7x^5y$)

(vii) $ab^2 - \frac{7}{a^2} + 5b^2 + 6$

- *Analysis:* The term $-\frac{7}{a^2}$ is equivalent to $-7a^{-2}$. Since the power of a is -2 (a negative integer), this is not a polynomial.
- **Is a Polynomial: No**